QUASI-FINITE PUSHFORWARD INDUCES A MORPHISM OF THE COH STACKS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 424-430 of sources/stacks-moduli/pinned-original-source.tex; proof span is lines 432-495, the complete author proof (62 body lines).

Editorial note. Navigation stubs: 12 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 3 of queue_moduli.json (pb=2835, stmt 424-430, proof 432-495).

Editorial note. stacks-moduli wave-1: fresh adjudication accept (2026-10-14); pool per Chief order 08:5x (dpa superseded to lane-s9); moduli.tex 53 labels/48 with proof judged 25 accept / 23 reject / 5 pb=0 env-dup rows excluded; spans cite-free and reference/slogan/history-free so no patchenv needed.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 07U8: sources/stacks-moduli/pinned-original-source.tex, source lines 495–495. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DK3: sources/stacks-moduli/pinned-original-source.tex, source lines 495–495. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0610: sources/stacks-moduli/pinned-original-source.tex, source lines 495–495. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0CZ5: sources/stacks-moduli/pinned-original-source.tex, source lines 495–495. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0CZ6: sources/stacks-moduli/pinned-original-source.tex, source lines 495–495. Cited by the retained statement or proof but not

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part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0CZ3: sources/stacks-moduli/pinned-original-source.tex, source lines 495–495. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0A4X: sources/stacks-moduli/pinned-original-source.tex, source lines 495–495. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 084Q: sources/stacks-moduli/pinned-original-source.tex, source lines 495–495. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 05WT: sources/stacks-moduli/pinned-original-source.tex, source lines 495–495. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DK1: sources/stacks-moduli/pinned-original-source.tex, source lines 495–495. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04NX: sources/stacks-moduli/pinned-original-source.tex, source lines 495–495. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0CZB: sources/stacks-moduli/pinned-original-source.tex, source lines 495–495. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let B be an algebraic space. Let $\pi : X \to Y$ be a quasi-finite morphism of algebraic spaces which are separated and of finite presentation over B. Then π_* induces a morphism $\mathit{Coh}_{X/B} \to \mathit{Coh}_{Y/B}$.*

Proof. Let $(T \to B, \mathcal{F})$ be an object of $\mathit{Coh}_{X/B}$. We claim

(a) $(T \to B, \pi_{T,*}\mathcal{F})$ is an object of $\mathit{Coh}_{Y/B}$ and
(b) for $T' \to T$ we have $\pi_{T',*}(X_{T'} \to X_T)^*\mathcal{F} = (Y_{T'} \to Y_T)^*\pi_{T,*}\mathcal{F}$.

Part (b) guarantees that this construction defines a functor $\mathit{Coh}_{X/B} \to \mathit{Coh}_{Y/B}$ as desired.

Let $i : Z \to X_T$ be the closed subspace cut out by the zeroth fitting ideal of $\mathcal{F}$ (Divisors on Spaces, Section 0CZ3). Then $Z \to B$ is proper by assumption (see Derived Categories of Spaces, Section 0CZB). On the other hand i is of finite presentation (Divisors on Spaces, Lemma 0CZ5 and Morphisms of Spaces, Lemma 084Q). There exists a quasi-coherent $\mathcal{O}_Z$-module $\mathcal{G}$ of finite type with $i_*\mathcal{G} = \mathcal{F}$ (Divisors on Spaces, Lemma 0CZ6). In fact $\mathcal{G}$ is of finite presentation as an $\mathcal{O}_Z$-module by Descent on Spaces, Lemma 0610. Observe that $\mathcal{G}$ is flat over B, for example because the stalks of $\mathcal{G}$ and $\mathcal{F}$ agree (Morphisms of Spaces, Lemma 0DK1). Observe that $\pi_T \circ i : Z \to Y_T$ is quasi-finite as a composition of quasi-finite morphisms and that $\pi_{T,*}\mathcal{F} = (\pi_T \circ i)_*\mathcal{G})$. Since i is affine, formation of i_* commutes with base change

(Cohomology of Spaces, Lemma 07U8). Therefore we may replace B by T, X by Z, $\mathcal{F}$ by $\mathcal{G}$, and Y by Y_T to reduce to the case discussed in the next paragraph.

Assume that $X \to B$ is proper. Then π is proper by Morphisms of Spaces, Lemma 04NX and hence finite by More on Morphisms of Spaces, Lemma 0A4X. Since a finite morphism is affine we see that (b) holds by Cohomology of Spaces, Lemma 07U8. On the other hand, π is of finite presentation by Morphisms of Spaces, Lemma 05WT. Thus $\pi_{T,*}\mathcal{F}$ is of finite presentation by Descent on Spaces, Lemma 0610. Finally, $\pi_{T,*}\mathcal{F}$ is flat over B for example by looking at stalks using Cohomology of Spaces, Lemma 0DK3. □